

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel International GCSE (9–1)

Friday 12 June 2026

Morning (Time: 1 hour 15 minutes)

Paper reference **4CH1/2C**

Chemistry
UNIT: 4CH1
PAPER: 2C

You must have:
 Calculator, ruler

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
 – *there may be more space than you need.*
- Show all the steps in the calculations and state the units.

Information

- The total mark for this paper is 70.
- The marks for **each** question are shown in brackets
 – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Write your answers neatly and in good English.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Cu and Cl have not been rounded to the nearest whole number.

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Answer ALL questions.

Some questions must be answered with a cross ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 This question is about polymerisation.

(a) The box gives some names used in polymerisation.

addition	alkane	alkene	condensation
dicarboxylic acid		poly(chloroethene)	water

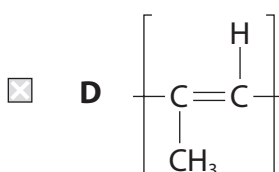
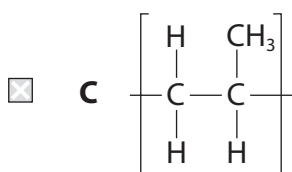
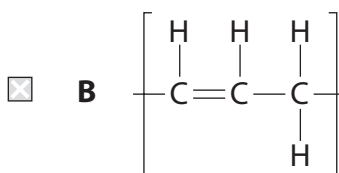
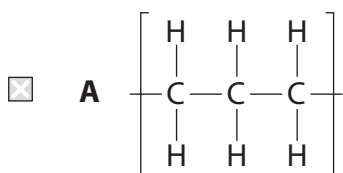
Complete the table by choosing the correct name from the box that matches each description.

(4)

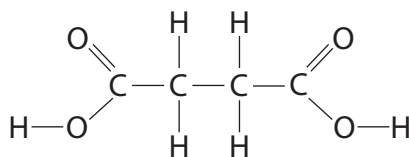
Description	Name
a monomer used to make an addition polymer	
a monomer used to make a condensation polymer	
a type of polymer that biodegrades	
a product of condensation polymerisation	

(b) Which structure shows the correct repeating unit for poly(propene)?

(1)



(c) This is the displayed formula for butane-1,4-dioic acid.

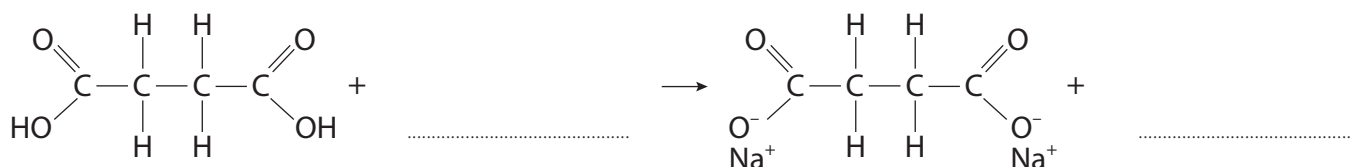


(i) Determine the molecular formula for butane-1,4-dioic acid.

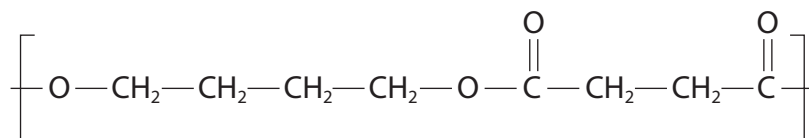
(1)

(ii) Complete the chemical equation for the reaction between butane-1,4-dioic acid and sodium.

(2)



(iii) Butane-1,4-dioic acid can react to form this polyester.



Draw a circle around the part of the polyester that comes from a butane-1,4-dioic acid molecule.

(2)

(Total for Question 1 = 10 marks)



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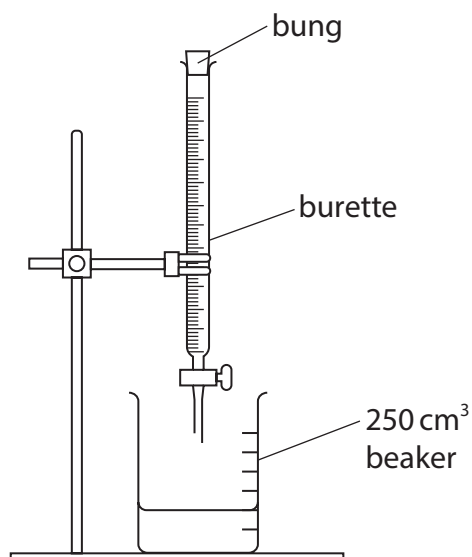


2 A titration can be used to prepare soluble salts.

(a) Name a negative ion that always gives a salt that is soluble in water.

(1)

(b) A student sets up this apparatus to do a titration.



The student makes two mistakes when setting up their apparatus.

Suggest two changes to the student's set up that need to be made if the titration is to be done correctly.

(2)

1

2

.....



(c) The student plans this method for their titration.

- add 25.0 cm^3 of an alkali to the beaker
- add an acid to the burette until the level reaches 0.00 cm^3
- open the burette tap and add acid to the beaker until there is a colour change
- record the final reading on the burette
- repeat the method until concordant results have been obtained.

Give three improvements to the student's method.

Do not refer to changes in apparatus in your answer.

(3)

1

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2

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3

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(d) Another student does a titration using the correct method.

The table shows their results.

Titration number	1	2	3	4
Volume of acid added in cm^3	22.95	22.65	23.05	22.90
Concordant results				

(i) Concordant results are those within 0.20 cm^3 of each other.

Add ticks (✓) to the table to show the concordant results.

(1)



(ii) Use the concordant results to calculate the mean volume of acid added.

(2)

mean volume = cm³

(e) Sodium hydroxide solution is used in titrations.

(i) Explain whether sodium hydroxide solution is an acid, an alkali or a salt.

(2)

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(ii) Solid sodium hydroxide is dissolved in excess water to make a solution.

This process is exothermic, with 47 kJ of energy released per mole.

Calculate the energy released, in kJ, when 0.55 kg of solid sodium hydroxide is dissolved in excess water.

[for sodium hydroxide $M_r = 40$]

(3)

energy released = kJ

(Total for Question 2 = 14 marks)

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3 This question is about electrolysis.

(a) Compounds that only contain covalent bonds cannot be electrolysed.

(i) Describe what is meant by the term **covalent bond**.

(2)

(ii) State why covalent compounds do not conduct electricity.

(1)

(b) Lead(II) bromide is an ionic compound that can be electrolysed.

(i) During electrolysis, the lead(II) bromide is at a temperature of 400 °C.

Explain a suitable material for the electrodes.

(2)

(ii) The lead(II) ions in this electrolysis have a positive charge.

Give the term used to refer to all positive ions.

(1)

(iii) Give the half-equation for the reaction at the positive electrode.

(1)

(iv) Identify the particles that flow through the wires connecting the electrodes.

(1)



(c) Sodium chloride can also be electrolysed.

- (i) The table gives the products at the positive electrode for the electrolysis of sodium chloride under different conditions.

Complete the table by giving the missing information.

(2)

Condition	Product at negative electrode	Product at positive electrode
molten sodium chloride		chlorine
aqueous sodium chloride		chlorine

- (ii) Explain how chlorine gas forms at the positive electrode.

(3)

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- (iii) Molten sodium chloride is electrolysed to give 5000 mol of chlorine gas.

Calculate the maximum volume, in dm^3 , of chlorine gas produced at rtp.

[molar volume of chlorine at rtp = $24\,000\text{ cm}^3$]

(2)

maximum volume = dm^3

(Total for Question 3 = 15 marks)

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4 Potassium chloride is a soluble salt with the formula KCl

(a) A flame test is done on a sample of potassium chloride.

State the colour of the flame.

(1)

(b) (i) Give a reason why adding blue litmus paper to a solution of potassium chloride is not a suitable test for chloride ions.

(1)

(ii) Give a suitable test for chloride ions.

(2)



(c) A student is given a sample of solid potassium chloride.

Describe a method that the student could use to make 250 cm^3 of 0.500 mol/dm^3 potassium chloride solution.

Include suitable calculations and apparatus in your answer.

[for potassium chloride $M_r = 74.5$]

(6)

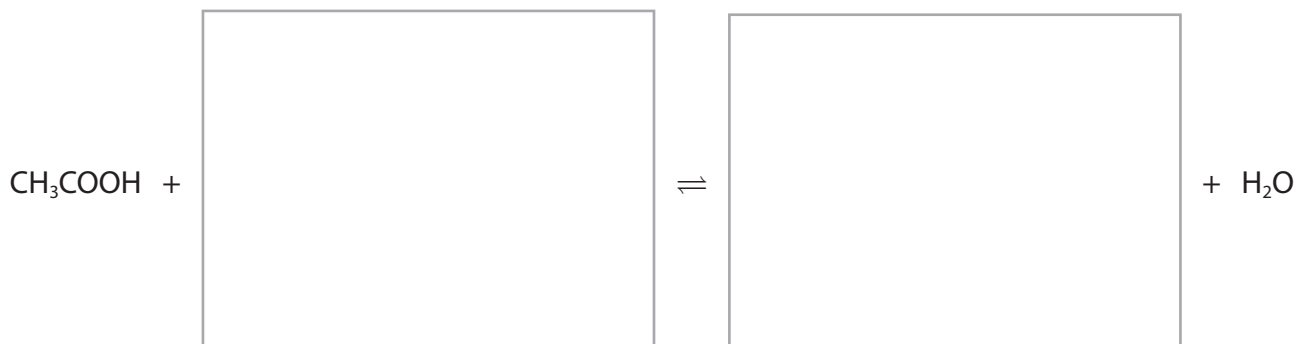
(Total for Question 4 = 10 marks)



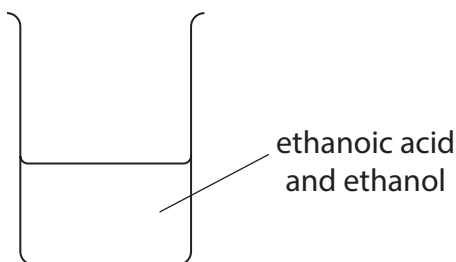
- 5 Ethanoic acid reacts with ethanol to produce ethyl ethanoate and water. This reaction is reversible.

(a) Complete the equation for the reaction between ethanoic acid and ethanol by adding the displayed formula of the missing reactant and the missing product.

(2)



(b) Ethanoic acid and ethanol are mixed in a beaker.



Give a reason why this reaction is unlikely to reach equilibrium.

(1)

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(c) The reaction between ethanoic acid and ethanol can be catalysed by sulfuric acid.

(i) Explain how a catalyst works.

(2)

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Give a reason for your answer.

(2)

Explain the effect that increasing the temperature would have on

- yield of ethyl ethanoate at equilibrium
- rate at which equilibrium is reached.

(5)

(Total for Question 5 = 12 marks)



6 This question is about the solubility of potassium sulfate in water.

(a) Give the formula for potassium sulfate.

(1)

This method can be used to determine the solubility of potassium sulfate at different temperatures.

- add a known mass of potassium sulfate to 25 cm³ of water in a boiling tube
- warm the boiling tube until all the potassium sulfate has dissolved
- cool the solution until crystals first appear, and record the temperature of the solution

Repeat the method with different masses of potassium sulfate.

(b) Give a suitable piece of equipment to measure the temperature of the solution.

(1)

(c) State the independent variable in this method.

(1)

(d) State why this method is not suitable for determining the solubility of barium sulfate.

(1)



- (e) The table shows the maximum mass of potassium sulfate that dissolves in 100 cm^3 of water at different temperatures.

Maximum mass of potassium sulfate in g	10.0	12.5	15.0	17.5	20.0
Temperature of water in $^{\circ}\text{C}$	12.7	27.4	41.6	55.6	69.5

- (i) Use data from the table to determine the mass of potassium sulfate dissolved in the boiling tube at 27.4°C .

(2)

mass = g

- (ii) A 200 cm^3 saturated solution of potassium sulfate is cooled from 55.6°C to 27.4°C .

Determine the mass, in g, of solid potassium sulfate that would crystallise.

(3)

mass = g

(Total for Question 6 = 9 marks)

TOTAL FOR PAPER = 70 MARKS



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